

Dataset: Intel Image Classification

Task: Six-Class Natural-Scene Image Classification

1. Abstract

This project presents a comparative study of three deep-learning approaches for six-class natural-scene image classification using the Intel Image Classification dataset. The investigated approaches are a Custom Convolutional Neural Network (CNN), ResNet18 using transfer learning, and a Vision Transformer (ViT). The project follows a complete workflow consisting of dataset exploration, preprocessing, train/validation splitting, data loading, model training, evaluation, and final comparison.

The dataset contains 14,034 training images and 3,000 test images distributed across buildings, forest, glacier, mountain, sea, and street classes. The training data were split into 11,227 training images and 2,807 validation images using an 80/20 split. The final comparison shows that the Vision Transformer achieved the highest test accuracy at 92.17%, followed by ResNet18 at 91.00% and the Custom CNN at 80.30%. The ViT also achieved the highest Macro F1 and Weighted F1 scores.

Keywords: deep learning, image classification, CNN, ResNet18, transfer learning, Vision Transformer, computer vision

2. Introduction

Image classification is a fundamental computer-vision task in which a model assigns an input image to one of a predefined set of categories. Deep-learning methods are widely used for this task because they can learn visual representations directly from image data.

This project focuses on natural-scene classification and compares three different architectural approaches. The Custom CNN provides a baseline built specifically for the project. ResNet18 represents pretrained convolutional transfer learning, while the Vision Transformer provides a transformer-based alternative. Comparing these approaches makes it possible to examine how model architecture affects classification performance.

3. Problem Statement

The problem addressed in this project is the automatic classification of natural-scene images into six predefined categories: buildings, forest, glacier, mountain, sea, and street. The main challenge is to develop and evaluate models that can distinguish visually different scene categories and generalize from the training data to unseen test images.

The project also investigates whether pretrained architectures provide better classification performance than a custom CNN baseline, and whether a transformer-based model can improve further on a strong convolutional transfer-learning model.

4. Project Objectives

The objectives of the project are:

- Explore and verify the image dataset and its class distribution.
- Prepare the images for deep-learning model input.
- Create a training and validation split and verify batched inputs.

Evaluate a Custom CNN baseline.

- Evaluate ResNet18 using transfer learning.
- Evaluate a Vision Transformer for the same classification task.

Compare the models using validation accuracy, test accuracy,

Macro F1, Weighted F1, and parameter counts.

5. Dataset Description

The project uses the Intel Image Classification dataset. The recorded dataset inspection identifies six scene categories. The training and test images are stored in class-specific folders and are loaded using torchvision's ImageFolder.

Class	Training	Test
Buildings	2,191	437
Forest	2,271	474
Glacier	2,404	553
Mountain	2,512	525
Sea	2,274	510
Street	2,382	501
Total	14,034	3,000

The class counts are not identical, but the distribution is reasonably balanced. This supports the use of both overall accuracy and class-sensitive F1 measures in the final evaluation.

6. Data Exploration and Preprocessing

The data-exploration stage verifies the dataset paths, folder structure, class names, training and test image counts, class distribution, sample images, and image properties. The recorded image inspection shows RGB images with an example size of 150×150 pixels before model-specific preprocessing.

For the recorded ViT pipeline, images are resized to 224×224 pixels and converted to tensors. The training data are divided into training and validation subsets using an 80/20 split. The recorded split contains 11,227 training images and 2,807 validation images. A batch size of 32 is used, and batch verification confirms tensors with shape $32 \times 3 \times 224 \times 224$.

7. Methodology

7.1 Custom CNN

The Custom CNN is used as the baseline model. Its purpose is to establish a reference performance level using a convolutional architecture built for the project. The final recorded test accuracy is 80.30%.

7.2 ResNet18 Transfer Learning

ResNet18 represents the transfer-learning approach. A pretrained convolutional architecture is used to provide stronger learned visual representations than the baseline model. The final recorded test accuracy is 91.00%.

7.3 Vision Transformer (ViT)

The ViT implementation uses the pretrained torchvision ViT-B/16 architecture. The pretrained backbone is frozen and the final classification head is replaced with a six-class linear layer. The recorded model

contains 85,803,270 total parameters and 4,614 trainable parameters. The recorded training run used five epochs and reached its best validation accuracy of 92.55% at epoch 3.

8. Experimental Setup

The project uses separate notebooks for data exploration, model training/evaluation, and final model comparison. The experimental workflow is organized as follows:

Stage	Description
Data exploration	Verify folders, classes, image counts, distribution, samples, and image properties.
Preprocessing	Resize images and convert them to tensors for model input.
Train/validation split	Use an 80/20 split of the training data.
DataLoaders	Create batched training, validation, and test inputs.
Model training	Train/evaluate the Custom CNN, ResNet18, and ViT.
Evaluation	Measure test accuracy and classification metrics.
Comparison	Compare performance and parameter counts.

9. Results and Evaluation

The final comparison notebook records the following results for the three models:

Model	Best Validation	Test Accuracy	Macro F1	Weighted F1
Custom CNN	80.69%	80.30%	80.56%	80.43%
ResNet18	90.92%	91.00%	91.16%	90.97%
Vision Transformer	92.55%	92.17%	92.35%	92.12%

The test-accuracy ranking is Vision Transformer (92.17%), ResNet18 (91.00%), and Custom CNN (80.30%). The Vision Transformer improves on the Custom CNN by 11.87 percentage points and on ResNet18 by 1.17 percentage points.

10. Model Comparison

The Custom CNN provides the lowest test accuracy and F1 scores and therefore serves primarily as the baseline. ResNet18 shows a substantial improvement, demonstrating the benefit of the pretrained convolutional approach. The Vision Transformer achieves the highest test accuracy and the highest Macro and Weighted F1 scores.

The parameter comparison also reveals a performance-efficiency trade-off. The recorded parameter counts are:

Model	Total Parameters	Trainable Parameters
Custom CNN	13,235,270	13,235,270

ResNet18	11,179,590	3,078
Vision Transformer	85,803,270	4,614

ViT gives the strongest predictive performance, but its total parameter count is much larger. ResNet18 provides a close test accuracy with far fewer trainable parameters, making it a strong alternative when model efficiency is important.

11. Conclusion

This project completed a full deep-learning image-classification workflow using the Intel Image Classification dataset. The work included data exploration, preprocessing, training/validation preparation, model development, evaluation, and final comparison.

Among the three evaluated approaches, the Vision Transformer achieved the strongest overall predictive performance, with 92.17% test accuracy, 92.35% Macro F1, and 92.12% Weighted F1. ResNet18 achieved 91.00% test accuracy and represents a strong alternative with substantially fewer trainable parameters. The Custom CNN achieved 80.30% and provided the baseline.

Final conclusion: The Vision Transformer is the best-performing model in this experiment, while ResNet18 offers a strong balance between classification performance and trainable model size.

12. References

Kaggle.IntelImageClassification. Datasetby PuneetBansal.

[1] Intel Image Classification dataset, as used in the project notebooks.

<https://www.kaggle.com/datasets/puneet6060/intel-image-classification?resource=download>

14. Project Organization

The project is organized into separate notebooks so that each stage can be inspected independently. The data-exploration notebook covers dataset paths, dataset information, class distribution, sample images, image properties, preprocessing, the train/validation split, DataLoaders, and batch verification. The model notebooks contain the corresponding training and evaluation workflows, while the final model-comparison notebook consolidates the final results and produces the comparison tables and figures.

This organization keeps data preparation, model development, and final analysis clearly separated and makes the project easier to review.